"The institute, in print"

The imsi exchange

Contents

Latest News 1
Thesis Defenses at IMSI : April 2026 . 1
Science Day at IMSI - April 16, 2026 1
In Memoriam: A Tribute to Pr. Ahmed Ben Ayad 2
Can Factories Think? If Yes, Who Controls Whom? 2
Between Circuits, Personalities, and Machines 2
From Idea to Execution: Our First Hackathon Experience 3
The challenge That Changed Physics 4
PySimVerse : A Python Drone Programming Simulator 4

Latest News

Thesis Defenses at IMSI : April 2026

The month of April saw a rich series of doctoral defenses at our institute. The IMSI Exchange congratulates the new doctors and presents a summary of their work.

Defense 1: Dr. BOUZANA Abdelghani

Title: Modeling of friction in spur gears

Supervisor: Pr. BELARIFI Farid

Abstract: This work investigates the tribological behavior of gears under elastohydrodynamic lubrication (EHD). Using ANSYS Fluent for numerical modeling, the study analyzes lubricant flow, contact pressures, and temperature distribution to predict wear mechanisms including surface fatigue, abrasion, and adhesion. The findings contribute to optimizing gear geometry, lubricant selection, and maintenance strategies for improved energy efficiency and system longevity.

Keywords: Gears, Tribology, Elastohydrodynamic lubrication, Wear, Numerical modeling, ANSYS Fluent

Defense 2: Dr. BELARBI Farah

Title: Development of a Fire Detection Method Using Image Processing Techniques

Supervisor: Pr. HASSINI Abdelatif

Abstract: Over the last decade, the number of fires has increased in industries, buildings, urban areas, and forests, leading to serious consequences, including human losses, material damage, as well as environmental, social, and economic impacts. In this context, fire prevention and early detection represent major challenges for the protection of human lives and infrastructure. An effective solution to fight fires relies on the use of early warning systems that enable rapid detection of the affected areas before the fire becomes uncontrollable, thereby reducing its spread and minimizing risks.

With the rapid development of the Internet of Things (IoT), surveillance cameras are now widely deployed, which has encouraged researchers to develop intelligent fire detection systems. These systems aim to detect fires as soon as they appear, to trigger an alarm and send alerts to safety officials. Deep learning has become an effective method for image classification and object detection. Convolutional neural networks (CNNs), in particular, are one of the most advanced approaches for fire detection, by improving detection accuracy. However, most existing research relies on very deep CNN architectures that are computationally expensive, require large memory capacity, and demand powerful hardware.

To address these gaps, this thesis proposes four new simple and lightweight CNN networks to classify and detect fires at an early stage: CNN_A, CNN_B, CNN_C, and CNN_D. Furthermore, in order to improve the algorithm's performance and reduce the error rate, a new ensemble approach is applied, combining the predictions of the four unique architectures. The results are evaluated using several metrics: accuracy, sensitivity, precision, F1-score, false positive rate (FP%) and false negative rate (FN%). In addition, the computational complexity of the models is analyzed, according to model size (in MB), detection speed (in frames per second – FPS), and the number of floating-point operations (MFLOPS).

A comparison of the proposed algorithms with recent fire detection approaches shows the effectiveness of the proposed approach. Among the evaluated models, the proposed ensemble model CNN_BC achieved the best results, with excellent accuracy of 99.69% and 96.01% on two challenging test sets, while presenting a reduced number of floating-point operations and a smaller size. Therefore, the CNN_BC model stands out for its robustness and its ability to detect fires at an early stage.

Keywords: Fire detection, Image classification, CNN, Surveillance system, Computer vision, Ensemble learning

Defense 3: Dr. ABERKANE Amira

Title: Treatment and Valorization of Plastic Waste for Sustainable Environmental Protection through the Development of Innovative Composite Materials.

Supervisor: Pr. BENDAIKHA LEBSIR Hayet

Abstract: This study investigates the use of untreated natural fibers from *Atriplex halimus* as reinforcement in recycled thermoplastic composites based on polypropylene (rPP) and high-density polyethylene (rHDPE). The fibers were incorporated at varying contents (5%, 10%, and 15%) using compression molding.

The results showed that rPP retained its mechanical strength better, even at higher fiber loadings, while rHDPE exhibited a more significant improvement in stiffness, particularly at 15% fiber content. The optimal balance between strength and stiffness was observed at 5% fiber content, with good interfacial adhesion between the matrix and the fibers-especially in the rPP-based composites, as confirmed by scanning electron microscopy (SEM) analysis.

The promising performance achieved without any chemical treatment of the fibers highlights the potential of *Atriplex halimus* as an eco-friendly and cost-effective reinforcement. This approach adds value to both an underutilized plant resource and recycled plastics, positioning the resulting composite material as a sustainable and efficient alternative for environmentally conscious applications.

Keywords: *Atriplex halimus*, natural fibers, recycled thermoplastics, polypropylene (rPP), high-density polyethylene (rHDPE), mechanical properties, tensile strength, stiffness, SEM analysis, sustainable composites.

Defense 4: Dr. MATIB Miloud

Title: Integration of the Ergonomic Dimension in the Optimization of Technical Systems: A Means of Accident Risk Prevention in Technological Systems.

Supervisor: Pr. LOUINS Zoubida

Abstract: The rapid evolution of technologies and the increasing complexity of systems in the oil and gas industry make ergonomics essential in the design of modern facilities. When neglected, issues such as poorly designed workstations, complex human–machine interfaces, and unsuitable working environments lead to fatigue, stress, human errors, and an increased risk of accidents, severely affecting productivity and safety. Our study examines these challenges and the level of understanding of ergonomics among designers and engineers, particularly the factors that hinder its integration into new designs. By exploring best practices and the obstacles encountered, our aim is to demonstrate how well-designed ergonomics improves working conditions and operational performance while minimizing risks.

Our work is structured around a literature review, an analysis of accident cases related to ergonomic deficiencies, and field surveys, in order to propose recommendations for integrating ergonomics from the design phase. We thus hope to provide a holistic and proactive perspective for safer, more productive installations adapted to the challenges of this complex sector.

Keywords: Prevention, Design, Ergonomics, Human factors, Accident.

The IMSI Exchange extends its warmest congratulations to all graduates and wishes them success in their future endeavors.

Science Day at IMSI - April 16, 2026

The Institute of Maintenance and Industrial Safety, in coordination with the Maintenance Department, organized a Study Day on Science Day.

The program included opening remarks by Pr. Chennoufi (Institute Director), Pr. Aissani (Vice-Director), and Pr. Benfkir (Head of Industrial Engineering). Workshops on **Python** (led by Pr. Mouffok) and **Matlab** skills (led by Pr. Moulai-khatir) were offered to the attendance.

The event concluded with a (rich) discussion between the attendance parts.

The tribute below was written in collaboration with Pr. D.E. Metahri, who knew Pr. Ben Ayad closely and generously offered time, reflections, and guidance to ensure this piece honors his memory faithfully.

In Memoriam: A Tribute to Pr. Ahmed Ben Ayad

The IMSI community mourns the loss of a distinguished figure who shaped generations of engineers and researchers.

With the passing of Pr. Ahmed Ben Ayad on April 18, 2026, the academic community of the University of Oran remembers one of its distinguished pillars and a symbol of dedication and academic excellence.

The late Pr. Ben Ayad was a professor and former Director of the Institute of Maintenance and Industrial Safety. He devoted his life to educating generations of students, instilling in them knowledge, discipline, and a strong passion for their profession. His remarkable academic career was also marked by his role as Vice-Rector in charge of pedagogy.

He was widely respected for his professionalism, wisdom, and humility, leaving a lasting impact on his colleagues, students, and all those who had the privilege of knowing him.

At the IMSI Exchange, we join the wider university community in honoring his memory. His name will remain present in our institute, and his contributions stand as a testament to a life dedicated to teaching, commitment, and service.

May he rest in peace.

A. Rahou
1st year industrial engineering student

Can Factories Think? If Yes, Who Controls Whom?

Human... the weakest link in the industrial system

This statement may sound harsh, but it reflects a reality that is quietly taking shape. In the age of artificial intelligence, speed of processing and decision accuracy are no longer in favor of humans. While we still cling to our role as controllers, factories have begun to learn how to operate without us... and perhaps even better than us. Cognitive manufacturing is not merely a technological evolution; it is a direct challenge to the idea of "human control." Factories capable of thinking, self-learning, and making critical decisions within fractions of a second-without waiting for any human intervention. This transformation raises an uncomfortable question: Are we developing these systems to serve us... or are we training a replacement for ourselves? In the near future, the question may no longer be how many jobs will disappear, but rather: will humans still have an essential role inside factories at all?

From Rigid Systems to "Living" Systems: Were Factories Always Unintelligent?

Let us state the truth clearly: traditional factories were not intelligent... they were only "obedient."

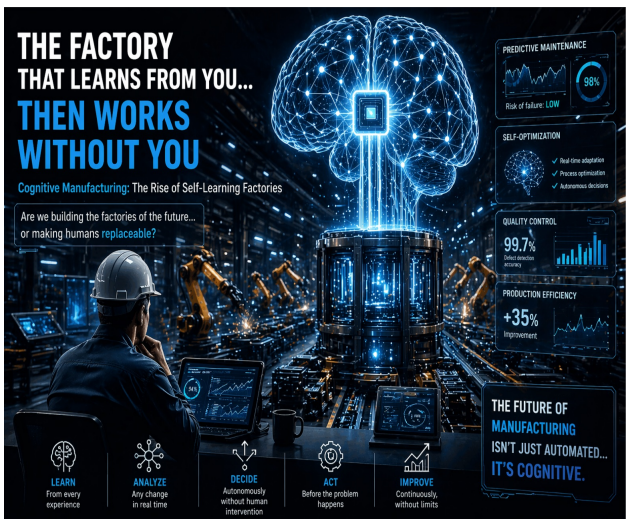
These factories rely on rigid structures and preprogrammed production plans, making them unable to adapt to sudden changes. In contrast, the human brain operates as a highly complex system that adapts in real time, containing approximately 86 billion neurons capable of continuously reorganizing themselves. Here emerges a controversial and striking concept: What if we no longer need systems that merely "execute"... but systems that "decide"?

Self-organizing industrial systems attempt to replicate this model, transforming factory components into a neural-like network that exchanges information and makes collective decisions without a central leader. The shocking question: Is this evolution... or the beginning of loss of control?

Perhaps the question is no longer: How do we make factories smarter? But rather: Should we make them so in the first place? Because the moment a factory becomes capable of independent learning and decision-making... may be the same moment when humans lose their monopoly over decision-making. And here remain several open questions:

- Can we trust the decisions of a fully autonomous system?
- What happens if the system makes a misjudgment?
- Will the role of engineers decrease or transform into supervisors of intelligent systems?

I do not believe the real challenge lies in our ability to build "thinking factories," but in our ability to coexist with them. The engineer learns to become a decision-maker within systems, while at the same time developing systems that no longer need his decisions.



In the future, factories may operate with higher efficiency and make faster and more precise decisions, but that does not necessarily mean we will fully understand them. And here arises the question: if we lose the ability to understand the systems we design, do we still control them... or are we simply watching them operate? We are between two choices: either we reinforce our role through awareness, or we find ourselves outside the equation we built with our own hands.

M. E. A. Ziati
1st year · Industrial Engineering · HR Leader, IEEE USTO Student Branch

Between Circuits, Personalities, and Machines

From a middle-school personality test to engineering, artificial intelligence, and the community that made it all make sense.

I. The Test That Took Root

It started, as many things do in middle school, with a trend. Everyone around me was suddenly obsessed with four letters - INTJ, ENFP, INFP - sharing their Myers-Briggs results like badges of identity. I joined in the way teenagers join most things: half out of curiosity, half because everyone else was doing it. But somewhere between reading about cognitive functions and staying up too late on psychology forums, something shifted. I stopped treating it as a game. I started treating it as a lens.

The Myers-Briggs Type Indicator is a personality framework built on Carl Gustav Jung's theory of psychological types - a theory that human behavior is not random, but follows predictable patterns rooted in how individuals prefer to perceive the world and make decisions. Jung identified two fundamental orientations (Introversion and Extraversion), two ways of gathering information (Sensing and Intuition), and two ways of making judgments (Thinking and Feeling). Isabel Briggs Myers and her mother Katharine Cook Briggs later added a fourth dimension - Judging versus Perceiving - producing the sixteen-type system we know today. It maps four dimensions of how a person thinks and engages with the world:

- **E - Energy:** Where do you direct your attention? Inward toward ideas and reflection, or outward toward people and action?
- **N-S - Perception:** How do you gather information? Through patterns and possibilities (Intuition), or through concrete facts and details (Sensing)?
- **T-F - Judgment:** How do you make decisions? Through logic and objective analysis (Thinking), or through values and interpersonal impact (Feeling)?
- **J-P - Structure:** How do you organize your world? With plans and closure (Judging), or with flexibility and open options (Perceiving)?

It is important to state clearly: MBTI is not a scientific law. It is a descriptive model - a map of tendencies, not a verdict on capability.

My Type - INTJ - The Architect

Introverted - Intuitive - Thinking - Judging. The INTJ mind builds systems. It looks past surface details toward underlying structures and long-term implications. INTJs are among the rarest types, making up roughly 2% of the population - typically independent, strategic, and driven by a need to understand the "why" behind things, which makes them naturally suited to engineering, research, and the analytical work that AI increasingly demands.

At thirteen, I did not fully understand what that meant. By the time I reached university, I did. Knowing I was an INTJ did not define me. But it told me something honest: I was wired to find the logic inside complex systems. It explained why I gravitate toward frameworks, why I find ambiguity uncomfortable unless I have converted it into a model, and why I would eventually choose to pursue both engineering and a leadership role in parallel. That system, at the time, was human personality. Years later, it would be artificial intelligence - and the skills, I discovered, turned out to be the same.

II. AI in 2026: A Different Kind of Mind

When I arrived at IMSI as a first-year student in Industrial Engineering - Automatism and Instrumentation Maintenance - I entered a new world. Not the buzzword version of AI, not the science-fiction version, but the real, working, deeply logical architecture underneath. When we speak of artificial intelligence today, we are no longer describing a technology on the horizon. It is here, embedded in our search engines, writing assistants, diagnostic systems, and increasingly, our engineering pipelines. The conversation has shifted from "will AI replace us?" to something more nuanced: how do we work alongside it?

> *I did not choose AI. AI was simply the next system I needed to understand - and once I started, I could not stop.*

I did not want to simply use AI tools. I wanted to understand how they reason, how they process ambiguity, how they arrive at conclusions. Large Language Models - the architecture behind most modern AI assistants - do not retrieve answers from a database. They predict, layer by layer, what the most coherent and contextually appropriate next token should be, based on patterns learned from vast amounts of human text. They have no understanding in the human sense. But they have something remarkably similar to cognitive style: consistent tendencies in how they handle ambiguity, structure arguments, and respond to framing.

This is exactly what studying MBTI had trained me to do with people. I had spent years learning to read cognitive patterns in humans. When I encountered AI, that habit transferred directly. The medium was different. The skill was the same. Once I saw that, I stopped being a passive user. I became an active interlocutor - and I became better at thinking for myself.

For students in technical fields like ours, the challenge is not learning to use these tools. It is learning to think critically about when, why, and how to trust them. That requires something no algorithm currently possesses: judgment, creativity, and self-awareness.

What I learned - and this is the part that changed how I work - is that each of these tools has a cognitive style. Claude tends toward careful, hedged reasoning. ChatGPT is confident and fast. NotebookLM is conservative and source-faithful. Understanding these tendencies lets you choose the right tool for the right task. You cannot prompt well what you do not understand. That insight, born from years of reading human personality types, made me a better engineer of AI before I ever wrote a line of code.

> *AI does not think the way we do - and that is precisely what makes it powerful. Our task is not to imitate it, but to complement it.*

III. IEEE: Where Theory Meets Community

But knowledge alone is a quiet thing. It was a conference that changed everything else. I attended an IEEE event, not entirely sure what to expect. What I found was a room full of people who asked the same kinds of questions I did - students, engineers, researchers, all of them curious, all of them building something. I discovered the IEEE Student Branch at USTO that day, and I did not leave.

I got involved, took on responsibility, and eventually became HR Department Leader - a role I did not anticipate but one that turned out to suit the part of me that MBTI had quietly been describing for years: someone who understands people as systems, who finds the logic in human dynamics the same way they find it in circuits. My work revolves around people: recruiting motivated students, organizing internal events, and fostering a culture where members feel both challenged and supported. It is also where I have encountered, in practice, everything personality science predicts about group dynamics. The INTJ plans the structure; the ENFP energizes the room; the ISTJ ensures nothing falls through the cracks. Knowing when to give an introvert space to think, when to challenge a J-type's rigidity, when to anchor a P-type's energy - it became as practical as any engineering skill. The best teams are not composed of people who think alike. They are composed of people who understand how each other thinks.

Through IEEE, I also became part of the DZ Young Leader experience - a reminder that ambition in Algeria is not scarce. It simply needs the right environment to grow. The conversations I had, the projects that took shape, the late discussions about technology and its place in our country: all of it became part of who I am as a student and as a person. IEEE, at its core, is a community of engineers committed to advancing technology for the benefit of humanity. Being part of it as a first-year student is a reminder that the work of learning never happens in isolation - it happens in community, in exchange, and in print.

Today, I am working on my own projects, at the intersection of engineering, AI, and human behavior. The kid who took a personality test because it was trending never disappeared - he just found bigger questions to ask. MBTI taught me to understand myself. AI taught me to think more rigorously. IEEE taught me that neither is enough without community.

The thread that runs through all of it - from a middle-school personality test to an engineering degree to a leadership role to the pages of this Exchange - is a single question: how do minds work? Human minds, artificial minds, collective minds. In an era where AI handles more and more cognitive tasks, self-knowledge may become the last truly irreplaceable human advantage. The question keeps getting bigger. I find that encouraging.

T. Merzouk
1st year industrial engineering student

From Idea to Execution: Our First Hackathon Experience

It was my first time participating in a hackathon, and I approached the experience with both excitement and curiosity. As a beginner, I deliberately chose to step out of my comfort zone and engage in this challenge. I participated alongside my teammate, Sarra Bousaid, and together we formed the youngest team in the competition.

The hackathon was organized by Muhandis Club at the National Polytechnic School and took place over three days, from April 12 to April 15, 2026. The objective was to design and develop a web or mobile application addressing challenges in the pharmaceutical sector by connecting suppliers, pharmacists, and patients. The event was held onsite, which created an immersive and collaborative working environment.

I represented Basmat Muhandis Club as a team leader. Due to limited availability of other members, our team consisted only of Sarra Bousaid and myself. We both worked collaboratively on the design phase, sharing creative decisions and shaping the visual and functional aspects of the project together. I handled the programming and technical implementation, while maintaining close coordination with Sarra throughout the development process. Despite our limited experience, we worked in a highly collaborative and motivating atmosphere, prioritizing learning and growth over competition. We also received guidance and support from fellow members of Basmat Muhandi Club, who helped us navigate certain technical tools and provided useful direction during the event.

One of the most challenging aspects for both Sarra and me was working on the database, as it was our first experience developing a complete web project. In addition, the time constraints added significant pressure throughout the hackathon. However, on the final presentation day, we were able to present our project confidently and received encouraging feedback from the judges, the school administration, and other participants.

This experience taught me the importance of building diverse teams with complementary skills. I also gained valuable insight from a keynote session delivered by one of the judges, who emphasized the hackathon theme and helped broaden my understanding of what it means to think like a programmer. One of the most meaningful moments for me was witnessing Sarra overcome her stage anxiety and present in front of an audience for the first time. Despite her natural shyness, she demonstrated remarkable courage, and I was genuinely proud of her progress. This experience also made me realize that my learning journey is still at its beginning and that there is much more to explore and master.

Overall, the hackathon was an enriching experience both technically and personally. We learned extensively while also sharing enjoyable moments, particularly during the design phase. The collaborative spirit, laughter, and shared effort made the experience truly memorable. The project we developed

was named PharmaFind.

I would definitely participate again, inshaAllah. Both Sarra Bousaid and I have decided to further develop our skills and prepare ourselves for future challenges. I would also like to encourage fellow students to step into new experiences without waiting for the "perfect moment," as growth comes through action and continuous learning. This experience has significantly shaped my perspective and strengthened my motivation to progress further.

Thank you Sarra.
Thank you Muhandis Club.

I. DROUICHE
Second year engineering student

The challenge That Changed Physics

> *"Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things."*
>
> Isaac Newton

On the 29th of January 1697, Isaac Newton returned home after a long day's work to find a letter waiting in his post. It contained a challenge set by the mathematician Johann Bernoulli for the finest minds in the world. It read: Given two points A and B (not vertically aligned), what is the curve along which a mass will descend, under gravity alone, in the shortest possible time? Common sense might suggest a straight line, after all, it's the shortest distance. But if you gently bend the ramp into an arch, the mass gains more acceleration. Even though it travels a longer path, it moves faster, and that extra speed can beat the straight line. This enchanting puzzle became known as the problem of fastest descent, or in Greek, the brachistochrone problem. For centuries, mathematicians had tried to solve it, yet no one had found the truly perfect answer.

At that time, Newton was no longer what you'd call an active physicist or mathematician. He was serving as Warden of the Mint, a high government position. When he saw the challenge, he was irked and wrote, "I do not love to be dunned and teased by foreigners about mathematical things". Still, the problem was so irresistibly teasing that he ended up spending the rest of the day and the whole night on it. By morning, he had arrived at a solution, something that had taken Bernoulli two full weeks. Newton submitted his work to the Philosophical Transactions, a famous scientific journal located in London. True to character, he didn't sign it, so no one could immediately tell whose work it was. Yet when Bernoulli read the anonymous paper, he recognised the genius behind it and elegantly declared, "I recognise the lion by his claw.". And while Newton dominated Bernoulli, in this instance it was Bernoulli who outshone Newton with a wonderfully clever solution of his own.

To solve this problem, Bernoulli drew inspiration from ancient philosophers who believed that light always takes the shortest path. However, when light travels from one medium to another, it bends in a particular way, forming what is known as Snell's law:

$$\frac{\sin\theta_1}{n} = \frac{\sin\theta_2}{n},$$

where n is a constant and θ_1, θ_2 are the angles the light beam makes with the normal in the first and second medium. A great mathematician named Pierre de Fermat later proved that even in this case, light still follows the shortest path, but instead of distance, it is time that is minimized. The constant n is actually related to the speed of light in each medium. Rewriting Snell's law gives:

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}.$$

Bernoulli's brilliant idea was to transform this mechanical problem into an optical one. Instead of a mass sliding from point A to B, he imagined a light beam traveling through many different media between A and B. The beam would bend each time it entered a new medium, and if we take infinitely many thin layers "m", the path becomes a continuous curve (as shown in the figure below). Since light always follows the path that minimizes time, that curve must be the solution to the brachistochrone problem. Applying Snell's law to this layered medium yields:

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2} = \ldots = \frac{\sin\theta_m}{v_m} = k,$$

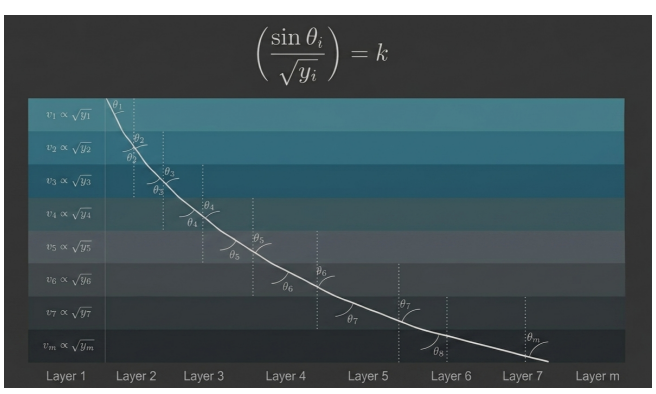
in other words : for all layers $i = 1, 2, \ldots, m$. That is, $\frac{\sin\theta_i}{v_i}$ is constant. Now consider energy conservation in the mechanical system. The mass starts from rest, and its potential energy is converted into kinetic energy. With the usual convention, $U + T = 0$, so $T = -U$. The kinetic energy is $\frac{1}{2}mv^2$ and the potential energy is mgy (where y is the vertical distance fallen). Thus:

$$\frac{1}{2}mv^2 = -mgy.$$

Rearranging, we find that the velocity is proportional to the square root of the vertical distance from the start: $v \propto \sqrt{y}$. Substituting into Snell's law gives:

$$\frac{\sin\theta_i}{\sqrt{y_i}} = k \text{ for all } 1 \leq i \leq m.$$

Bernoulli immediately recognized this as the equation of a cycloid. The angle θ_i can be expressed in terms of the rate of change of y . Substituting that into the above relation leads to a differential equation whose solution is indeed the cycloid. The cycloid is the path traced by a point on the rim of a rolling wheel. It has a remarkable property: no matter where you release the mass along the curve, it takes the same time to reach the lowest point. That is why it is also called the tautochrone curve, from the Greek for "same time."



Fascinated by his solution, Bernoulli wrote: "In this way I have solved at one stroke two important problems, an optical and a mechanical one, and have achieved more than I demanded from others. I had shown that the two problems, taken from entirely separate fields of mathematics, have the same character." Little did he know that he was onto something much greater. Because this solution will be the insight that inspired one of his students, Pierre Louis Maupertuis, to formulate one of the most important principles in physics: the principle of least action. This principle would go on to influence all of mechanics and physics, helping to reshape modern physics and leading to enormous discoveries in mathematics. Above all, it remains one of the greatest keys to unlocking a true understanding of the universe.

Science is elegant.

A. MOULAI-KHATIR
anes.mkh@gmail.com

PySimVerse : A Python Drone Programming Simulator

The IMSI Exchange invites readers to discover a new tool that may enrich their programming and simulation projects.

What Is PySimverse?

PySimverse is a lightweight Python simulator for drones, launched as a response to the discontinuation of the popular DJI Tello educational drone. It allows students, researchers, and developers to write Python code to control virtual drones - no physical hardware required.

Why the Exchange Recommends It

- **Accessibility:** Runs on a standard laptop - no expensive hardware needed.
- **Educational Value:** Ideal for self-learning, student projects, and club activities.
- **Scientific Rigor:** Flight time accuracy within 0.34 seconds of real-world flights.

Potential Applications at IMSI

Members of our community could explore PySimverse for:

- Student projects in automation and instrumentation.
- Club activities (Robotics Club, AI Club, IEEE Student Branch).
- Teaching support for courses on control systems or computer vision.
- Research prototypes before committing to physical hardware.

A Word from the Exchange

We encourage interested readers to explore the project, test the simulator, and share their experiences with the Exchange. Have you used PySimverse? Would you like to write a short review or tutorial for a future issue? Let us know at `imsiexchange@gmail.com`.